

INFORMATION THEORY & CODING

Dr. Bo Ning

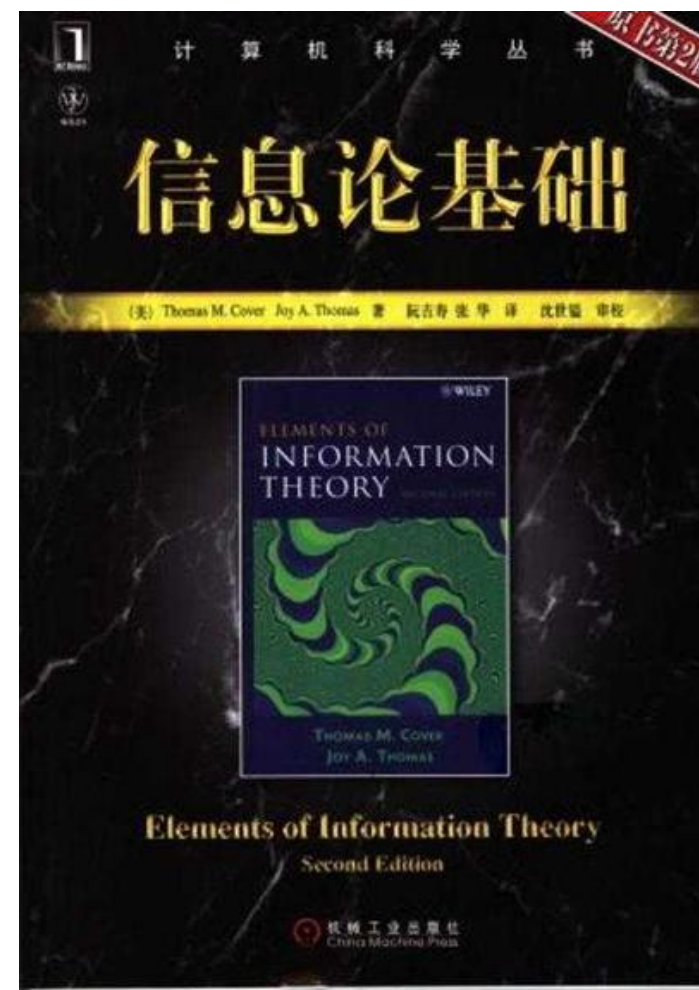
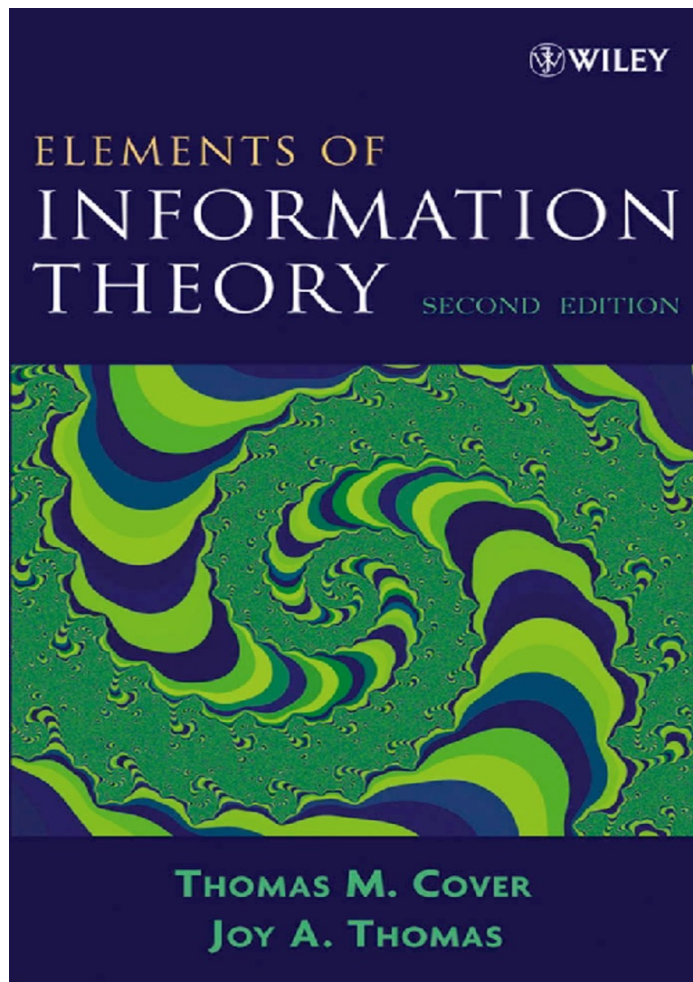
Associate Professor, College of Computer Science

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Textbooks and References

Textbooks

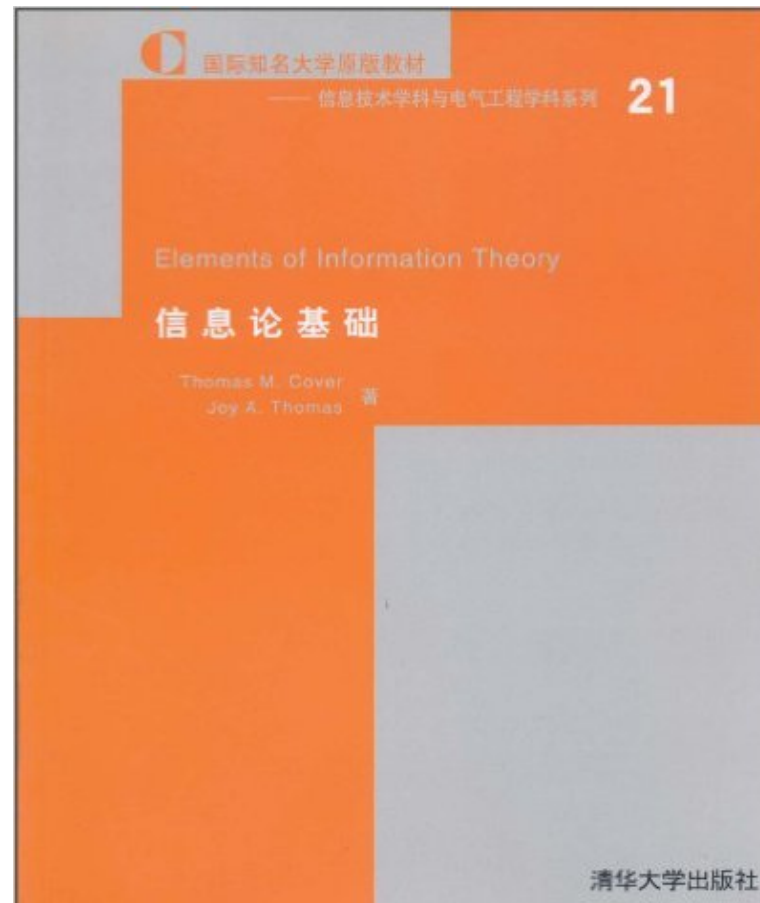
- Thomas M. Cover, Joy A. Thomas, [Elements of Information Theory](#), 2nd Edition, Wiley-Interscience, 2006.



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- Thomas M. Cover, Joy A. Thomas, [Elements of Information Theory](#), 1st Edition, Tsinghua University Press, 2003.



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References

- Raymond W. Yeung, [A First Course in Information Theory](#), Springer, 2002.
- F. J. MacWilliams, N.J.A. Sloane, [The Theory of Error-Correcting Codes](#), North-Holland, 1977.
- Shu Lin, D. J. Costello, [Error Control Coding](#), 2nd Edition, Principles of Mobile Communication, 2004.
- Original papers in [IEEE Transactions on Information Theory](#)

Schedule

Assessment

■ Homework assignments (? %)

- ◇ Assigned in class and posted on the course webpage
- ◇ Must be submitted **at the beginning of class** on due date
- ◇ No extension policy with exceptions

■ Project (? %)

- ◇ Select a related topic
- ◇ Could be a survey or a paper review, please submit a report + supporting materials

■ Final exam (? %)

- ◇ Covers the entire semester's material

Policy Reminders

Academic dishonesty consists of misrepresentation by deception or by other fraudulent means and can result in **serious consequences**, e.g. the grade of **zero** on an assignment, loss of credit with a notation on the transcript (“Grade of **F** assigned for academic dishonesty”).

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Note to the Reader:

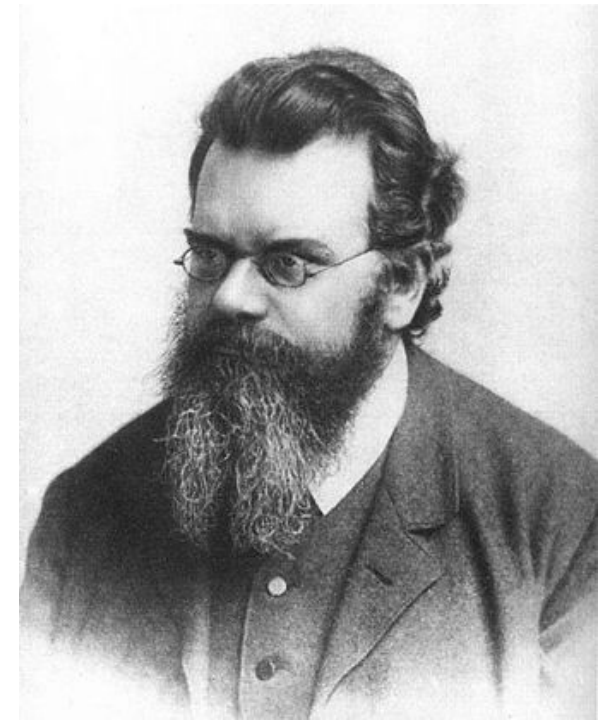
- We have drawn on a number of sources to produce these lecture notes.
- These lecture notes are a perpetual work in progress. Please report any typo or other errors by email. Thanks!
- We try to prepare these lecture notes carefully, but they are **NOT** intended to replace the textbook.
- For more information, please refer to bo.ning@nankai.edu.cn
- Office hours: drop by or appoint by email.

A Brief History *

1877 – Boltzmann showed that thermodynamic entropy is related to the statistical distribution of molecular configurations, with increasing entropy corresponding to increasing randomness.

$$S = k_B \log W$$

$$\text{where } W = N! \prod_i \frac{1}{N_i!}.$$



Ludwig Boltzmann
(1844-1906)

* For a more complete history, one may check the “Timeline of information theory” page from Wikipedia.

A Brief History

1924 – Nyquist rate and reconstruction of bandlimited signals from their samples. Also stated formula $R = K \log m$, where R is the rate of transmission, K is a measure of the number of symbols per second and m is the number of message amplitudes available. Amount of information that can be transmitted is proportional to the product of bandwidth and time of transmission.



Harry Nyquist
(1889-1976)

A Brief History

1928 – (inventor of the *oscillator*) – in the paper entitled “Transmission of Information” proposed formula $H = n \log s$, where H is the “information” of the message, s is the number of possible symbols, n is the length of the message in symbols.



Ralph V. L. Hartley
(1888-1970)

A Brief History

1938 – In his Master's thesis "*A Symbolic Analysis of Relay and Switching Circuits*" at MIT, he demonstrated that electrical application of Boolean algebra could construct and resolve any logical, numerical relationship.



Claude E. Shannon
(Apr. 30, 1916 – Feb. 24, 2001)

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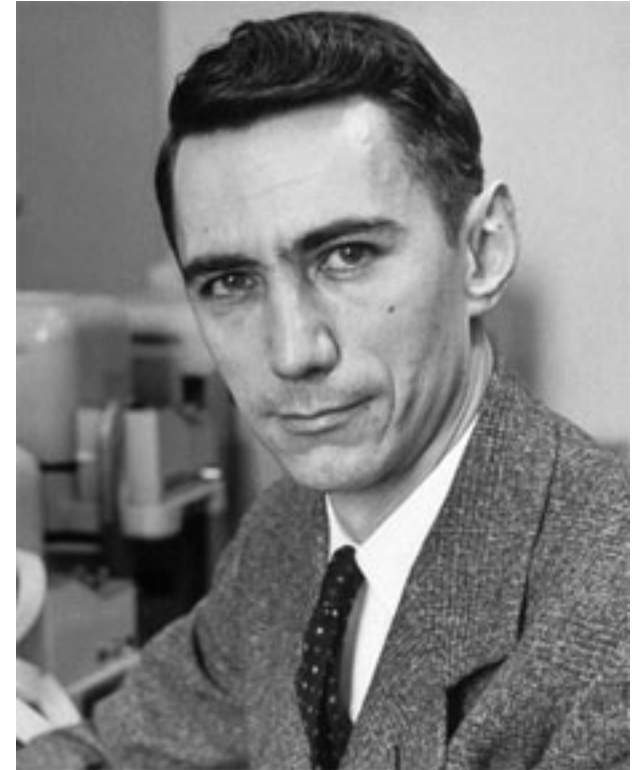
"possibly the most important, and also the most famous, master's thesis of the century."



Claude E. Shannon
(Apr. 30, 1916 – Feb. 24, 2001)

A Brief History

1948 – efficient source representation, reliable information transmission, digitalization – foundation of communication and information theory. Made the startling discovery that **arbitrarily reliable communications are possible at non-zero rates**. Prior to Shannon, it was believed that in order to get arbitrarily low probability of error, the transmission rate must go to zero. His paper “*A Mathematical Theory of Communications*” proved to be the foundation of **modern communication theory**.



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a
b
p
g
o
f

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

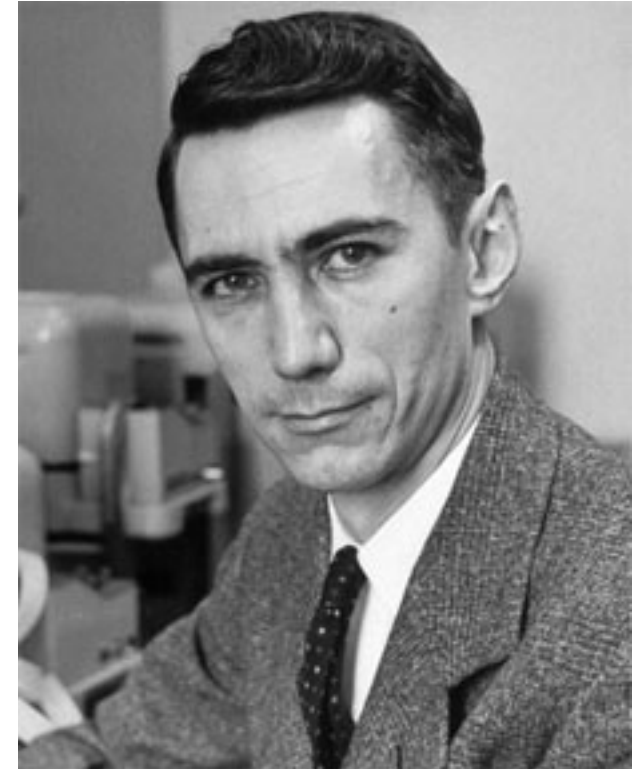
THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one *selected from a set* of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design.

If the number of messages in the set is finite then this number or any monotonic function of this number can be regarded as a measure of the information produced when one message is chosen from the set, all choices being equally likely. As was pointed out by Hartley the most natural choice is the logarithmic function. Although this definition must be generalized considerably when we consider the influence of the statistics of the message and when we have a continuous range of messages, we will in all cases use an essentially logarithmic measure.

The logarithmic measure is more convenient for various reasons:

t
y



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Shannon Centenary

Shannon Centenary

The Shannon Centenary

The *Shannon Centenary*, 2016, marks the life and influence of Claude Shannon on the hundredth anniversary of his birth on 30 April 1916. Shannon is best known for developing the mathematical foundations of communication (establishing the field of information theory) [1], data compression [2], digital computers [3], cryptography [4], circuit complexity [5], flow networks [6], and juggling [7], as well as laying foundations of artificial intelligence [8, 9] and human-computer interaction [10].

- A [video](#) that the Information Theory Society has created.
- [Shannon's Work and Its Legacy](#), M. Effros and H. V. Poor.
- David Forney's [comments on receiving the IEEE Medal of Honor](#).

www.itsoc.org/resources/Shannon-Centenary

Quotes

“What made possible, what induced the development of coding as a theory, and the development of very complicated codes, was Shannon’s Theorem: he told you that it could be done, so people tried to do it.”

– Robert Fano



“Before 1948, there was only the fuzziest idea of a message was. There was some rudimentary understanding of how to transmit a waveform and process a received waveform, but there was essentially no understanding of how to turn a message into a transmitted waveform.”

– Robert Gallager



Quotes

“ To make the chance of error as small as you wish? Nobody had ever thought of that. How he got that insight, how he even came to believe such a thing, I don't know. But almost all modern communication engineering is based on that work.”

– Robert Fano



A Brief History (cont')

1950 R. Hamming – Developed a family of **error-correcting codes**

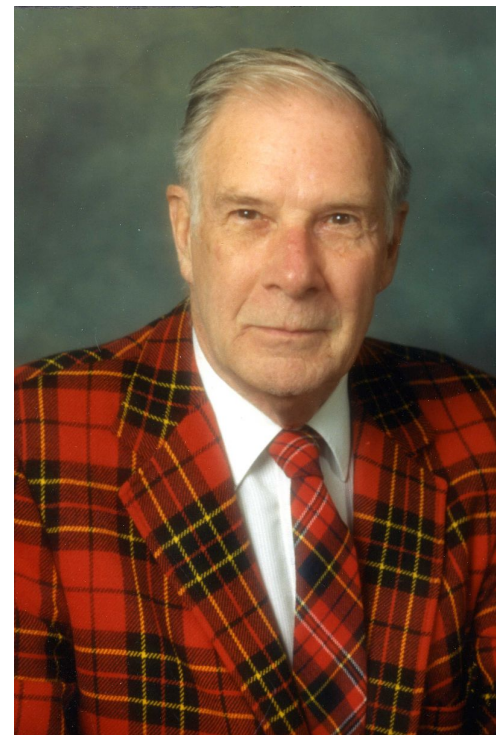
1952 D. Huffman – Efficient **source encoding**

1950-60's Muller, Reed, Solomon, Bose, Ray-Chaudhuri, Hocquenghem – **Algebraic Codes**

1970's Fano, Viterbi – **Convolutional Codes**

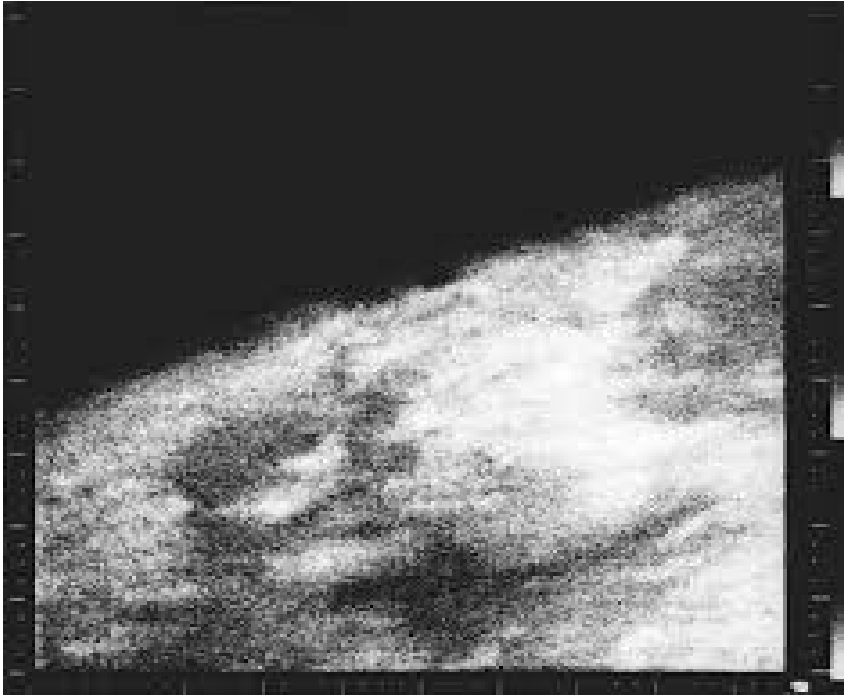
1990's Berrou, Glavieux, Gallager, Lin – Near capacity achieving coding schemes: **Turbo Codes, Low-Density Parity Check Codes**

2008 E. Arikan – First practical construction of codes achieving capacity for a wide array of channels: **Polar Codes**



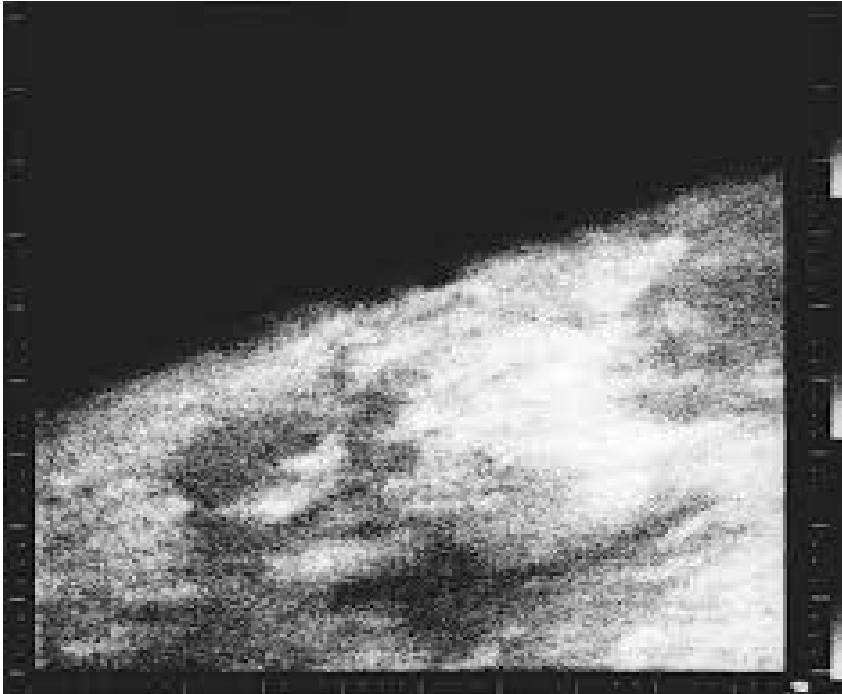
Richard W. Hamming
(1915-1998)

An example



Mars, Mariner IV, '64 using
no coding

An example

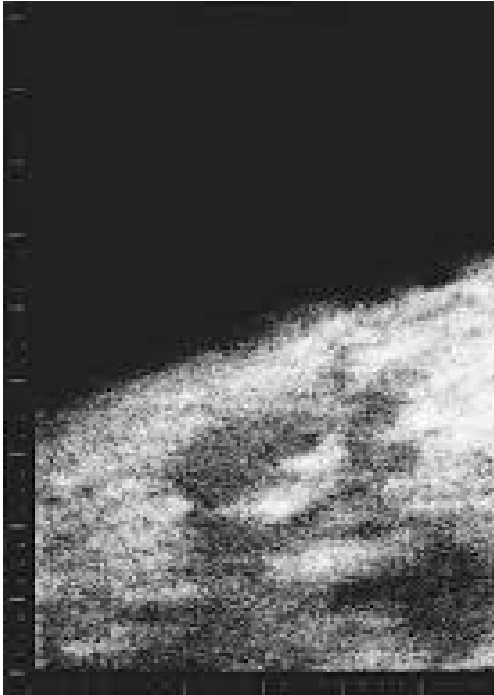


Mars, Mariner IV, '64 using
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Mars, Mariner VI, '69 using
Reed-Muller coding

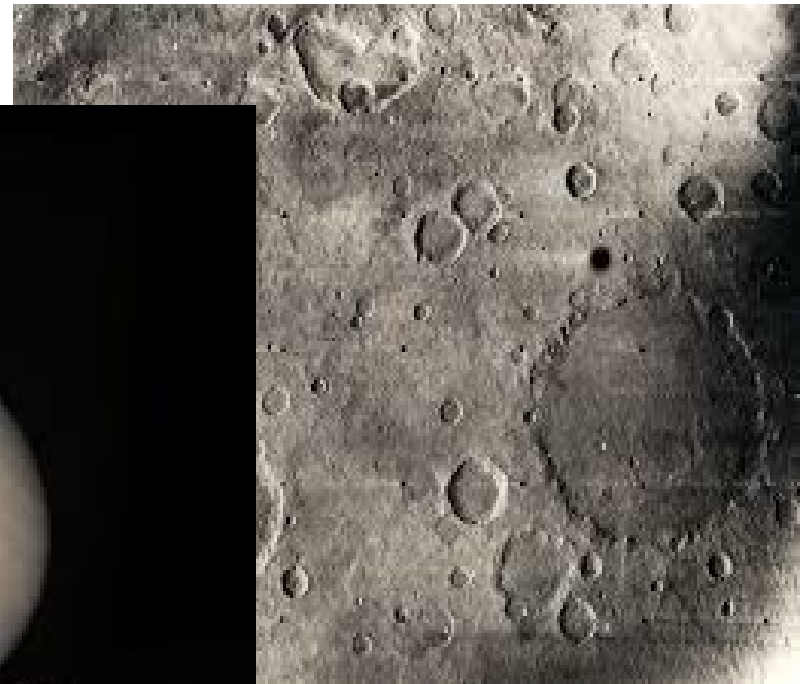
An example



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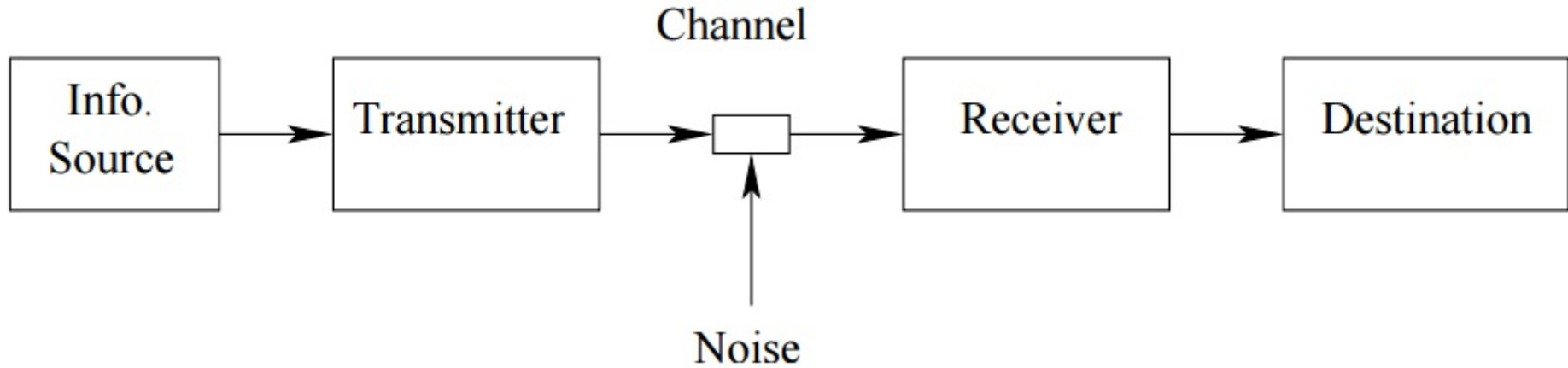


Saturn, Voyager, '71 using
Golay coding

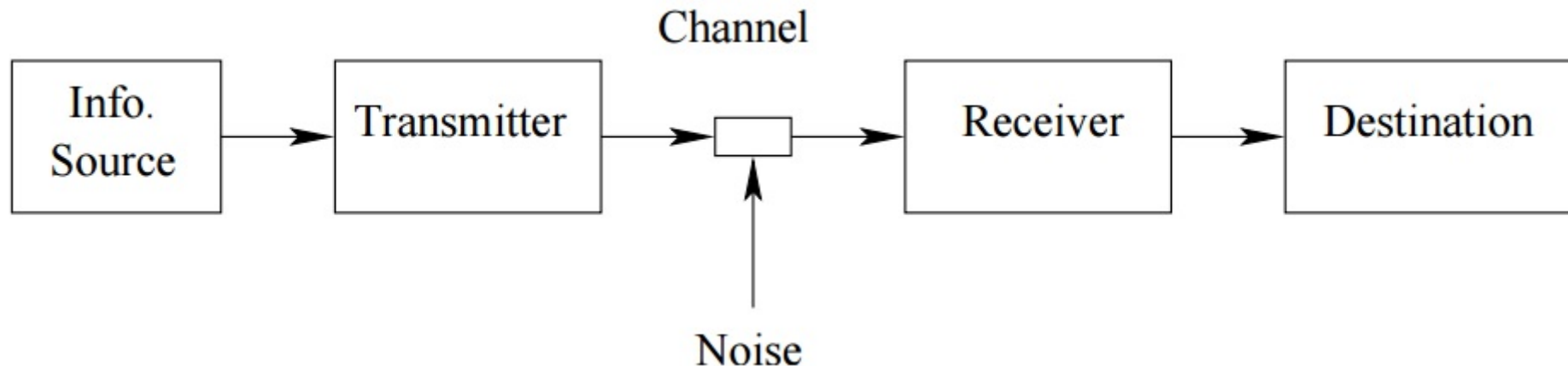


Mariner VI, '69 using
Golay coding

A Communication System

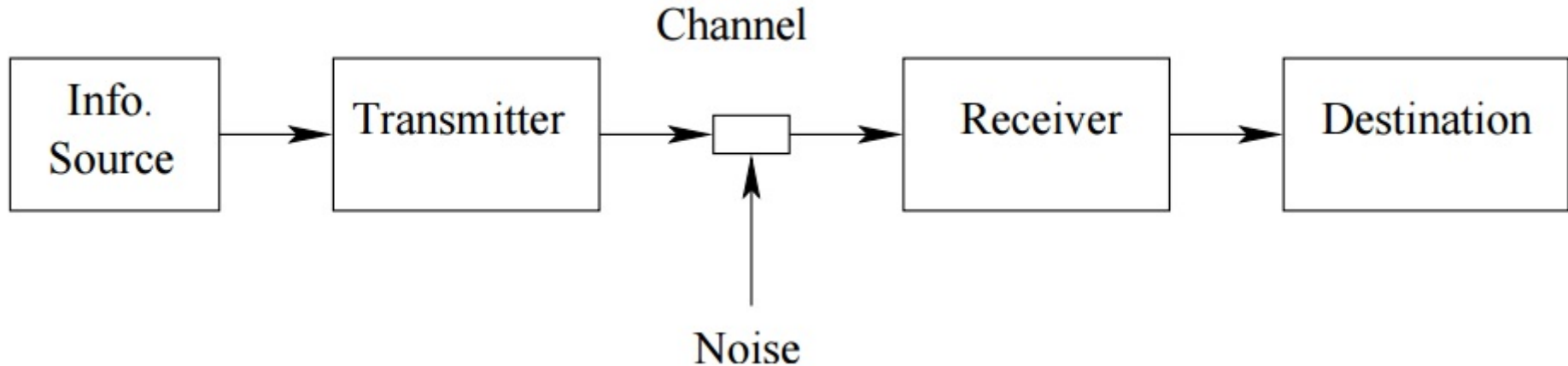


A Communication System



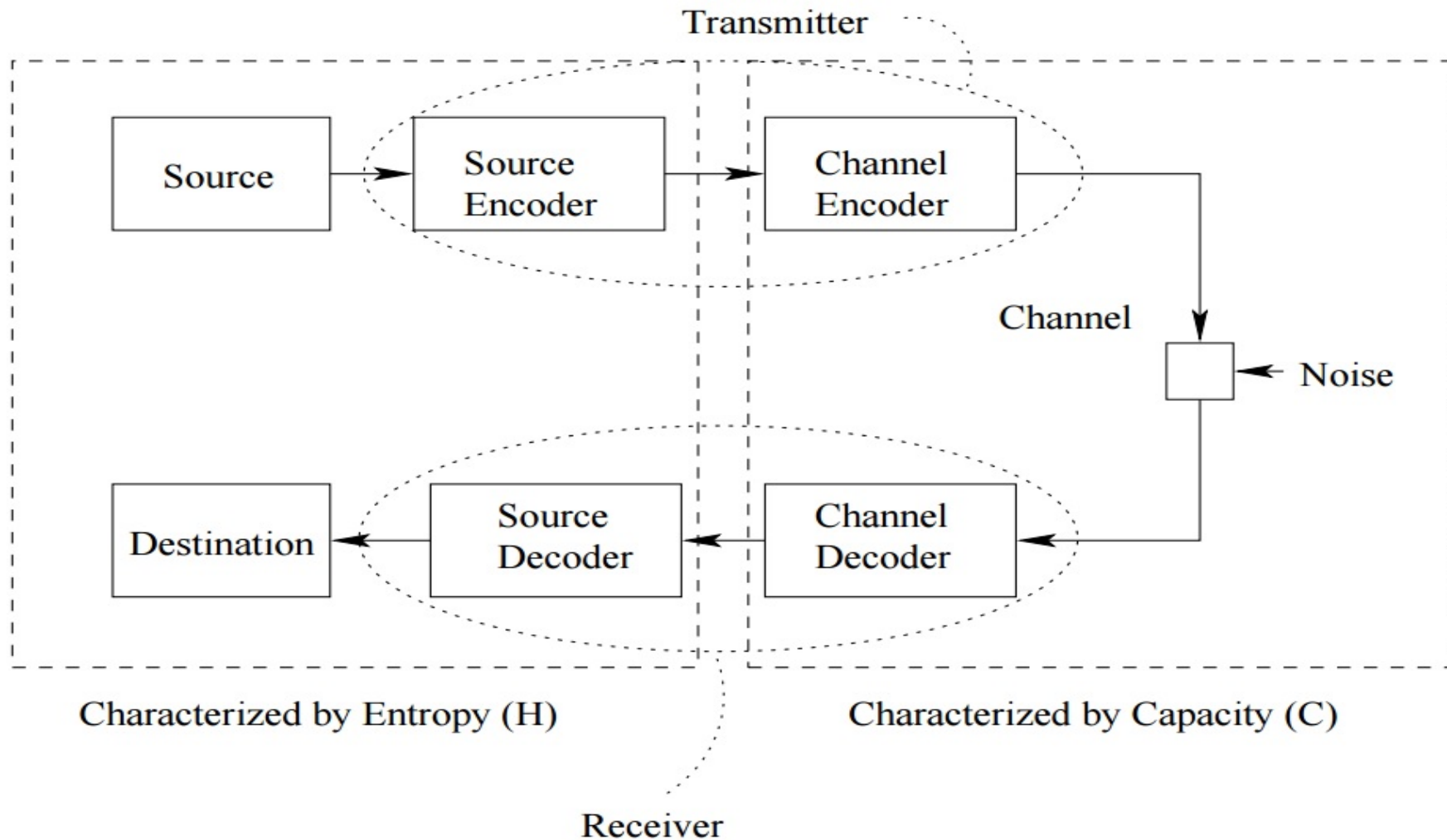
- **Info. Source**: any source of data we wish to transmit or store
- **Transmitter**: mapping data source to the channel alphabet in an efficient manner
- **Receiver**: mapping from channel to data to ensure “reliable” reception
- **Destination**: data sink

A Communication System



Question: Under what conditions can the output of the source be conveyed *reliably* to the destination? What is reliable? Low prob. of error? Low distortion?

An Expanded Communication System



What is the ultimate data compression (answer: the entropy H)?

What is the ultimate transmission rate of communication (answer: channel capacity C)?

Encoders

Source Encoder

- *map* from source to bits
- “matched” to the information source
- Goal: to get an *efficient* representation of the source (i.e., *least* number of bits per second, *minimum* distortion, etc.)

Encoders

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Channel Encoder

- *map* from bits to channel
- depends on channel available (*channel model*, bandwidth, noise, distortion, etc.) In communication theory, we work with *hypothetical channels* which in some way capture the *essential* features of the physical world.
- Goal: to get *reliable* communication

Source Encoder: Examples

- Goal: To get an efficient representation (i.e., **small** number of bits) of the source *on average*.

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Outcome	1	2	3	4	5	6	7	8
Representation	000	001	010	011	100	101	110	111

Answer: Require **3** bits to represent any given outcome.

Source Encoder: Examples

Example 2: Consider a horse race with 8 horses. It was determined that the probability of horse i winning is

$$\Pr[\text{horse } i \text{ wins}] = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64} \right)$$

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Answer 1: Let's try the code of the previous example.

Outcome	Probability	Representation 1
0	$\frac{1}{2}$	000
1	$\frac{1}{4}$	001
2	$\frac{1}{8}$	010
3	$\frac{1}{16}$	011
4	$\frac{1}{64}$	100
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To represent a given outcome, the average number of bits is $\bar{\ell} = 3$.

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Answer 2: What if we allow the length of each representation to vary amongst the outcomes, e.g., a Huffman code:

Outcome	Probability	Representation 2
0	$\frac{1}{2}$	0
1	$\frac{1}{4}$	10
2	$\frac{1}{8}$	110
3	$\frac{1}{16}$	1110
4	$\frac{1}{64}$	111100
5	$\frac{1}{64}$	111101
6	$\frac{1}{64}$	111110
7	$\frac{1}{64}$	111111

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5	$\frac{1}{64}$	111101
6	$\frac{1}{64}$	111110
7	$\frac{1}{64}$	111111

The average number of bits is

$$\begin{aligned}\bar{\ell} &= \frac{1}{2} + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{16} \cdot 4 + \frac{4}{64} \cdot 6 \\ &= 2\end{aligned}$$

Source Encoder: Examples

Definition: The source **entropy**, $H(X)$ of a random variable X with a probability mass function $p(x)$, is defined as

$$H(X) = \sum_x p(x) \log_2 \frac{1}{p(x)}$$

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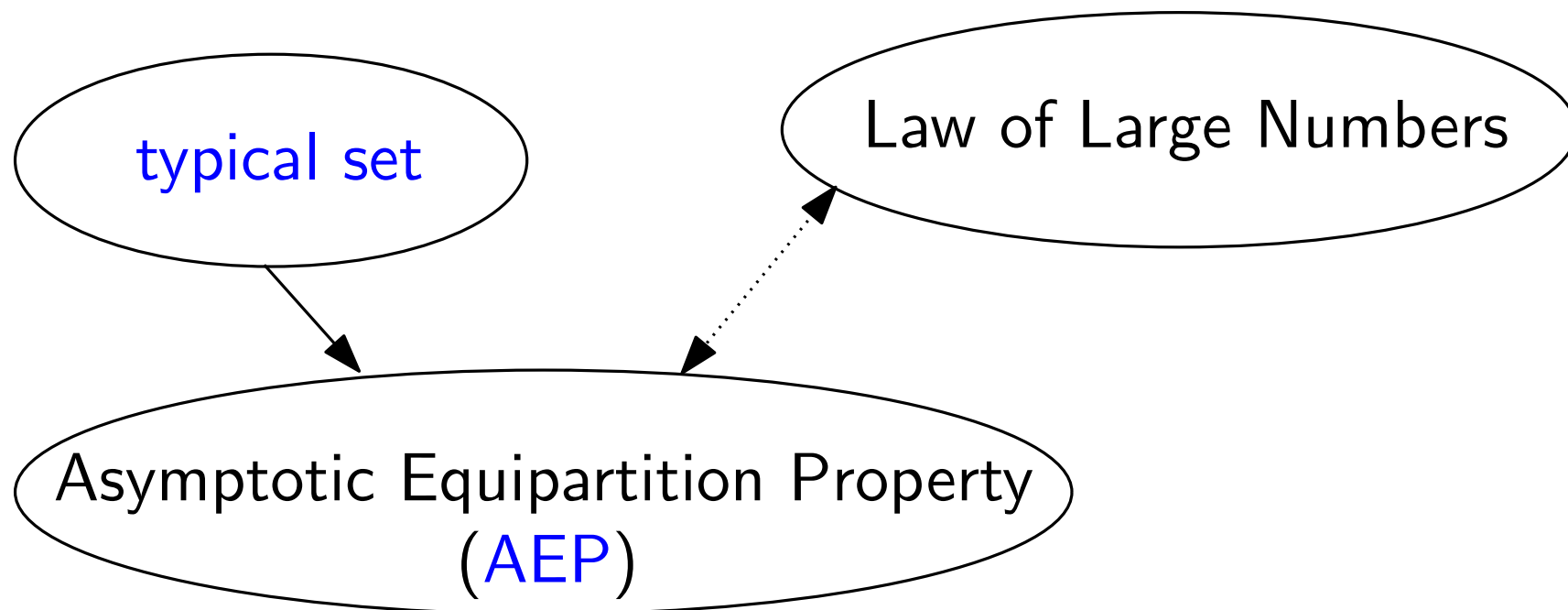
The Huffman code is **optimal**!

Source Encoder: Examples

- Information theory and coding deal with the “typical” or *expected* behavior of the source.
- Entropy is a measure of the *average* uncertainty associated with the source.

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With a channel code we add *redundancy* to the transmitted data sequence which allows for the correction of errors that are introduced by the channel.

Channel Encoder

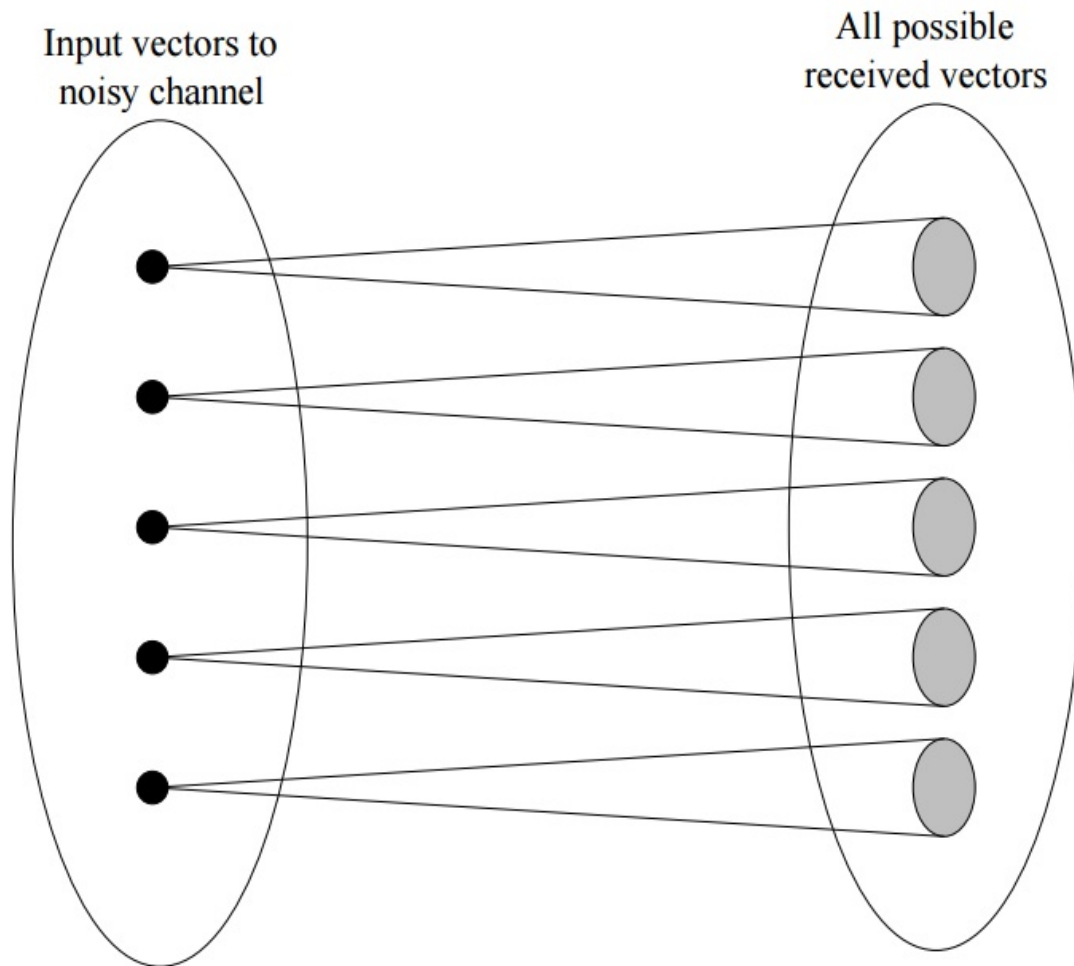
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Example:



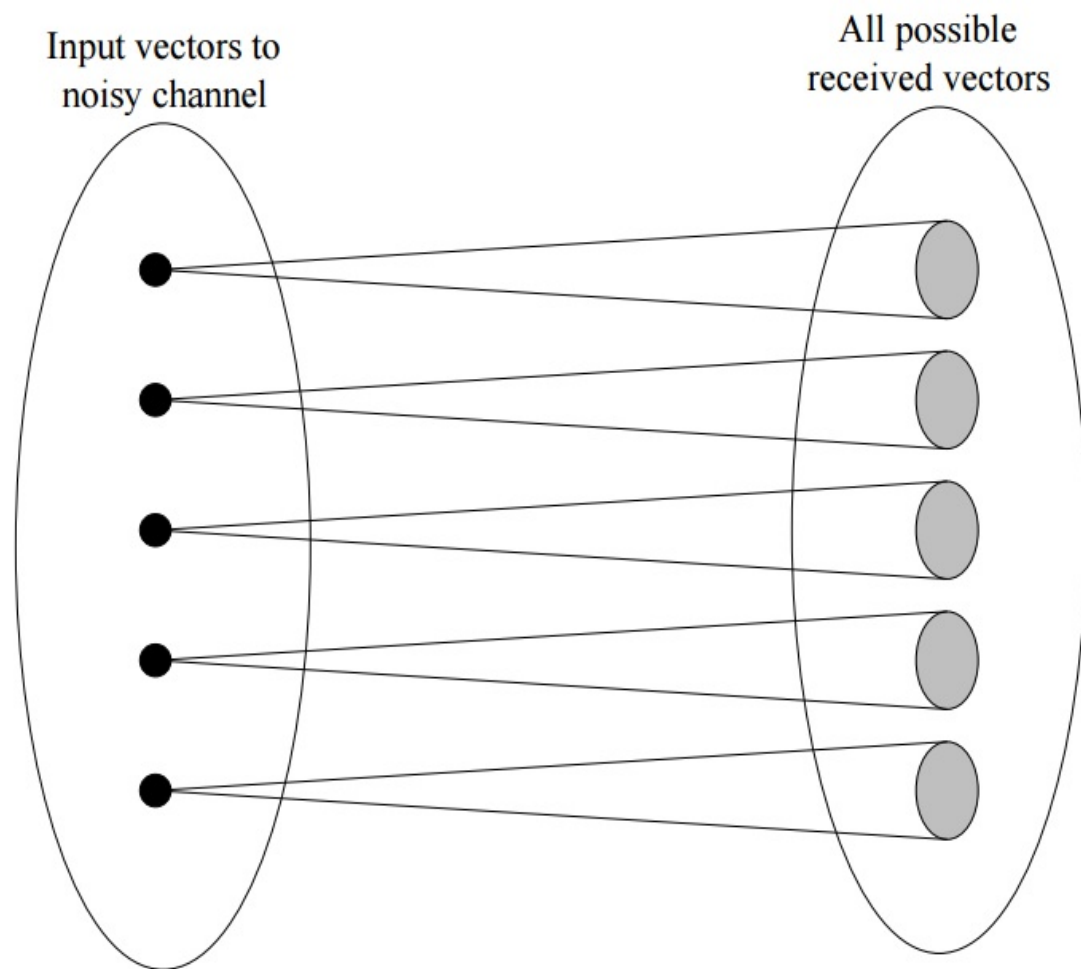
Channel Encoder



Each transmitted codeword is corrupted by the channel. Each codeword **corresponds to a set of possible received vectors**.

Specify a set of codewords so that at the receiver it is possible to **distinguish which element was sent with high-probability**.

Channel Encoder



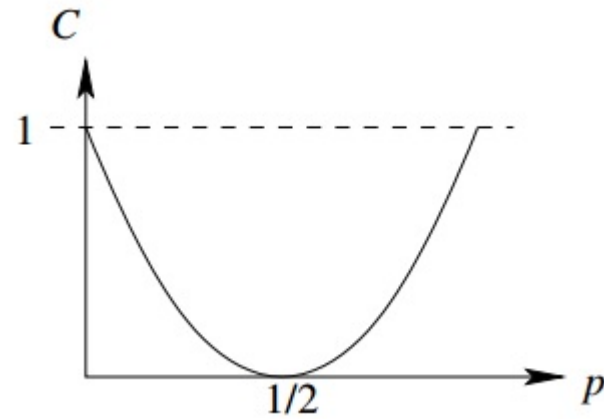
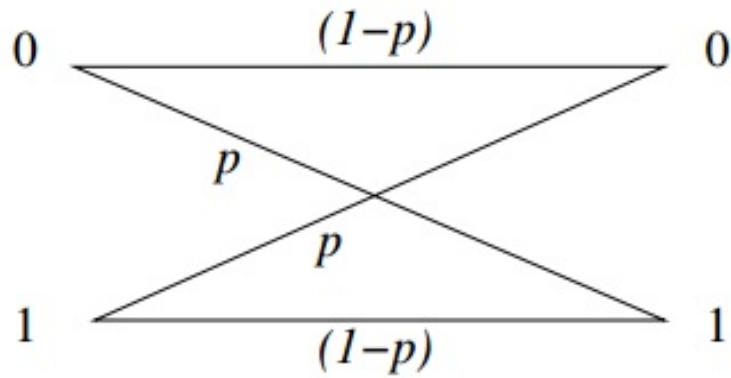
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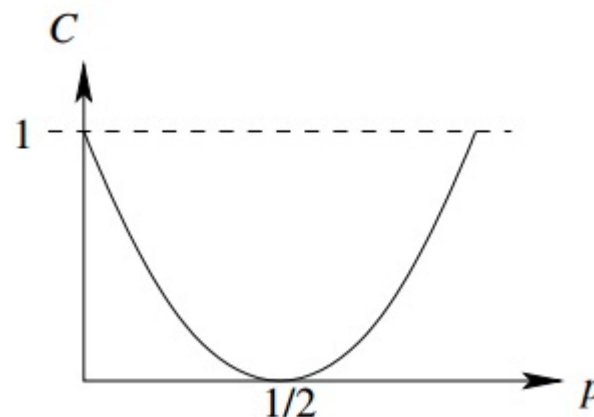
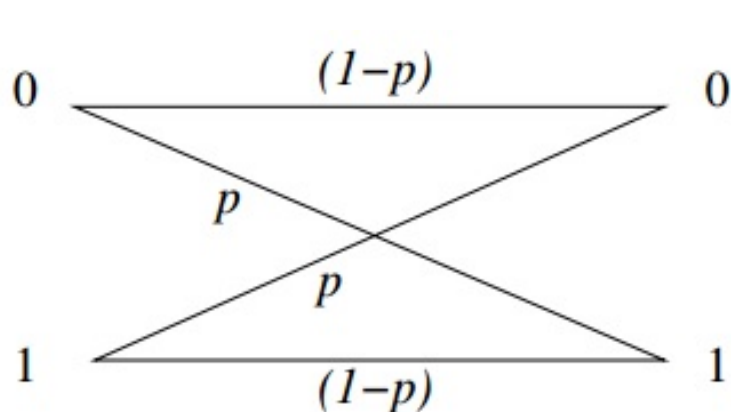
The **channel coding theorem** tells us the **maximum** number of such codewords we can define and still maintain completely distinguishable outputs.

Shannon's Channel Coding Theorem There is a quantity called the *capacity*, C , of a channel such that for every rate $R < C$ there exists a sequence of $(\underbrace{2^{nR}}_{\text{\#codewords}}, \underbrace{n}_{\text{\# chan. uses}})$ codes such that $\Pr[\text{error}] \rightarrow 0$ as $n \rightarrow \infty$. Conversely, for any code, if $\Pr[\text{error}] \rightarrow 0$ as $n \rightarrow \infty$ then $R \leq C$.

Example: Binary Symmetric Channel

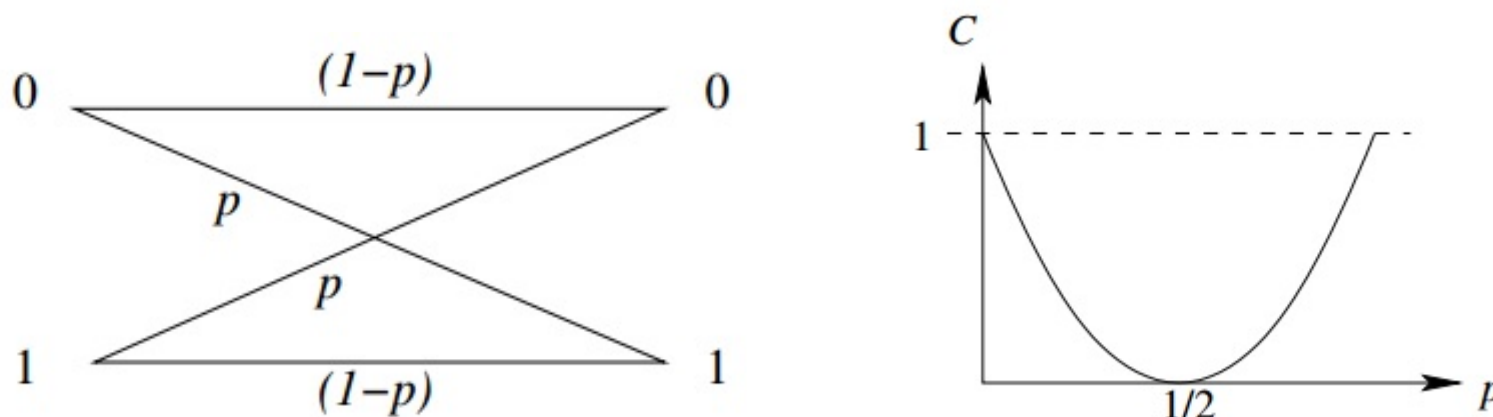


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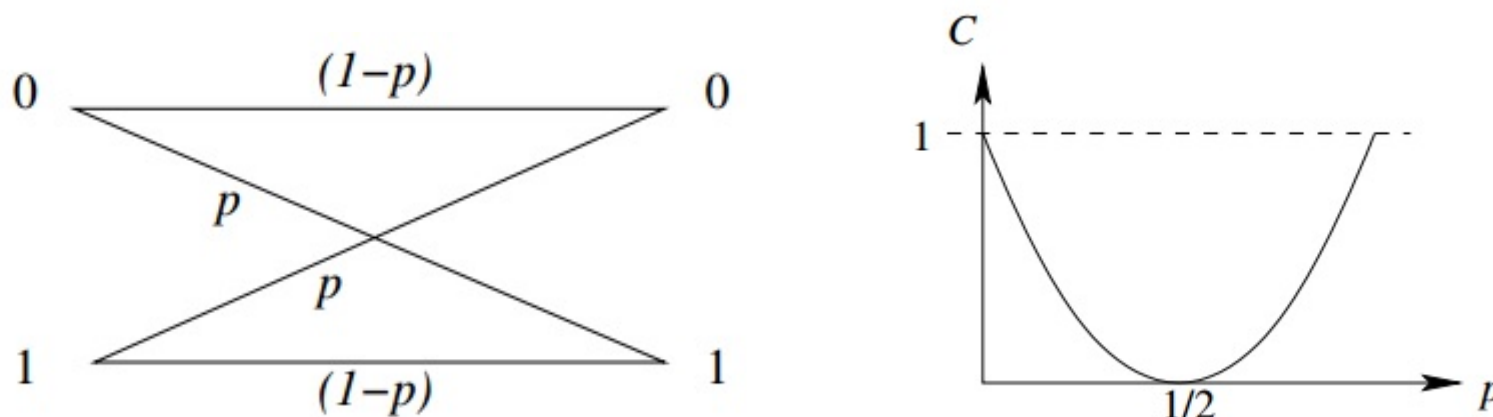
- Input channel alphabet = Output channel alphabet = $\{0, 1\}$
- Assume *independent* channel uses (i.e., memoryless)
- Channel randomly flips the bit *with probability p*
- For $p = 0$ or $p = 1$, $C = 1$ bits/channel use (*noiseless channel* or *inversion channel*)

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- **Worst case:** $p = 1/2$, in which case the input and the output are *statistically independent* ($C = 0$)
- **Question:** How do we *devise codes* which perform well on this channel?

Repetition Code

- In this code, we repeat one bit *odd times*. The code consists of *two* possible codewords:

$$\mathcal{C} = \{000 \cdots 0, 111 \cdots 1\}$$

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- Decoding by a *majority voting* scheme: if there are more 0's than 1's then declare 0, otherwise 1.
- Suppose that $R = 1/3$, i.e., the source output can be *encoded* before transmission by repeating each bit *three times*.

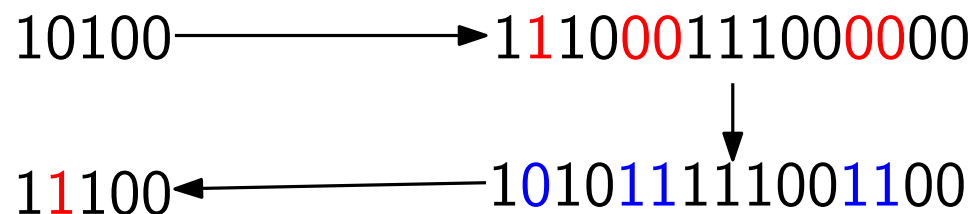
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$$\mathcal{C} = \{000 \cdots 0, 111 \cdots 1\}$$

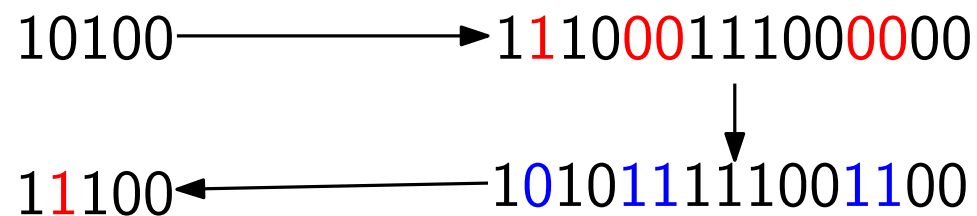
- Decoding by a *majority voting* scheme: if there are more 0's than 1's then declare 0, otherwise 1.
- Suppose that $R = 1/3$, i.e., the source output can be *encoded* before transmission by repeating each bit *three times*.

Example:



Repetition Code

Example:

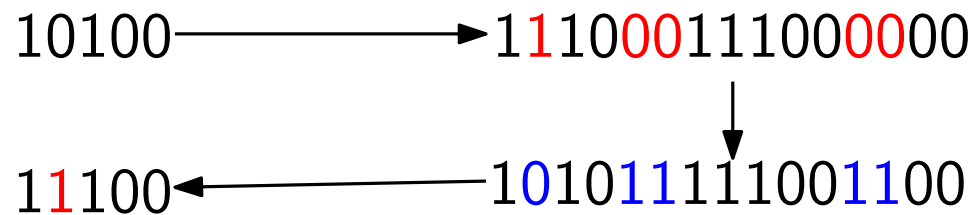


The *bit error probability* $\Pr_e$ is:

$$\begin{aligned}\Pr_e &= \Pr[2 \text{ channel errors}] + \Pr[3 \text{ channel errors}] \\ &= 3p^2(1 - p) + p^3 \\ &= 3p^2 - 2p^3.\end{aligned}$$

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If $p \leq 1/2$, $\Pr_e$ is less than p . So, the repetition code **improves** the channel's reliability. And for **small** p , the improvement is dramatic.

Repetition Code

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Repetition Code

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For $R = 1/(2m + 1)$, the *bit error probability* Pr_e is:

$$\begin{aligned}\text{Pr}_e &= \sum_{k=m+1}^{2m+1} \text{Pr}[k \text{ errors out of } 2m + 1 \text{ transmitted bits}] \\ &= \sum_{k=m+1}^{2m+1} \binom{2m+1}{k} p^k (1-p)^{2m+1-k} \\ &= \binom{2m+1}{m+1} p^{m+1} + \text{terms of higher degree in } p.\end{aligned}$$

Repetition Code

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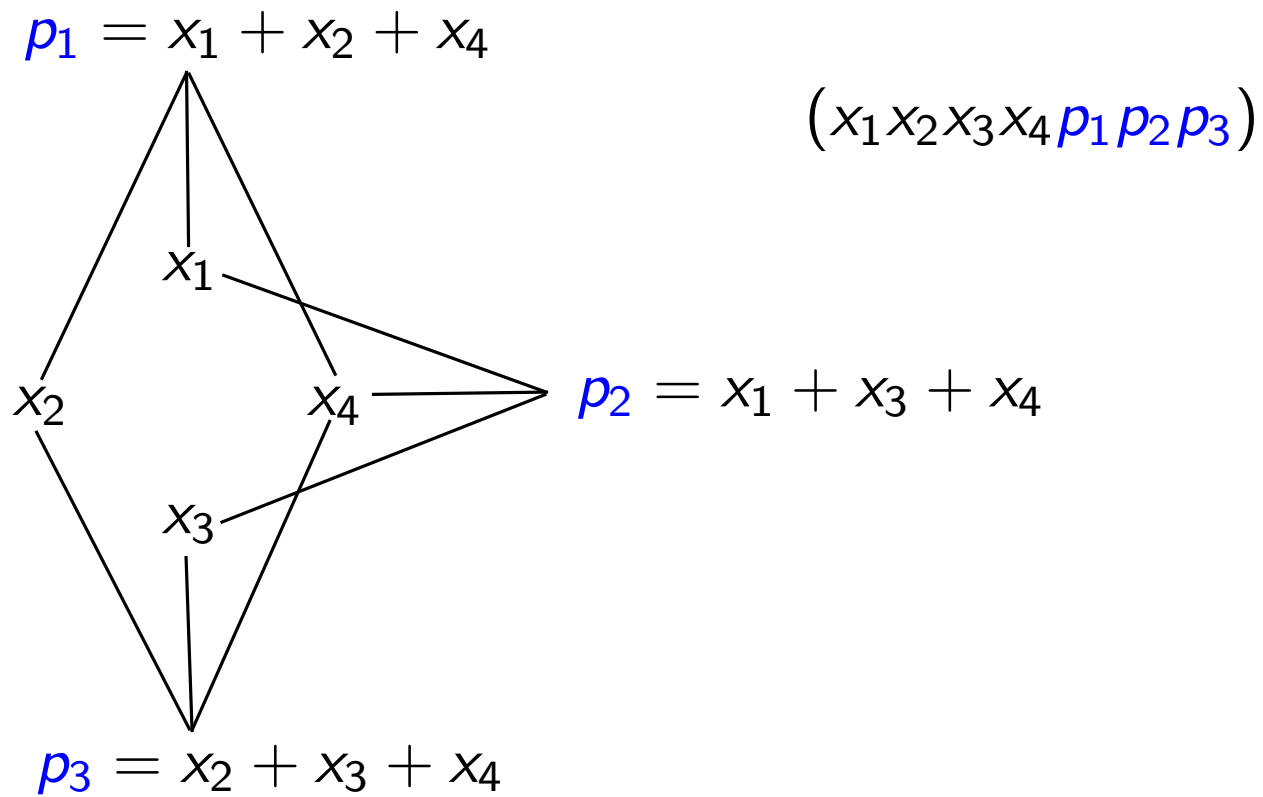
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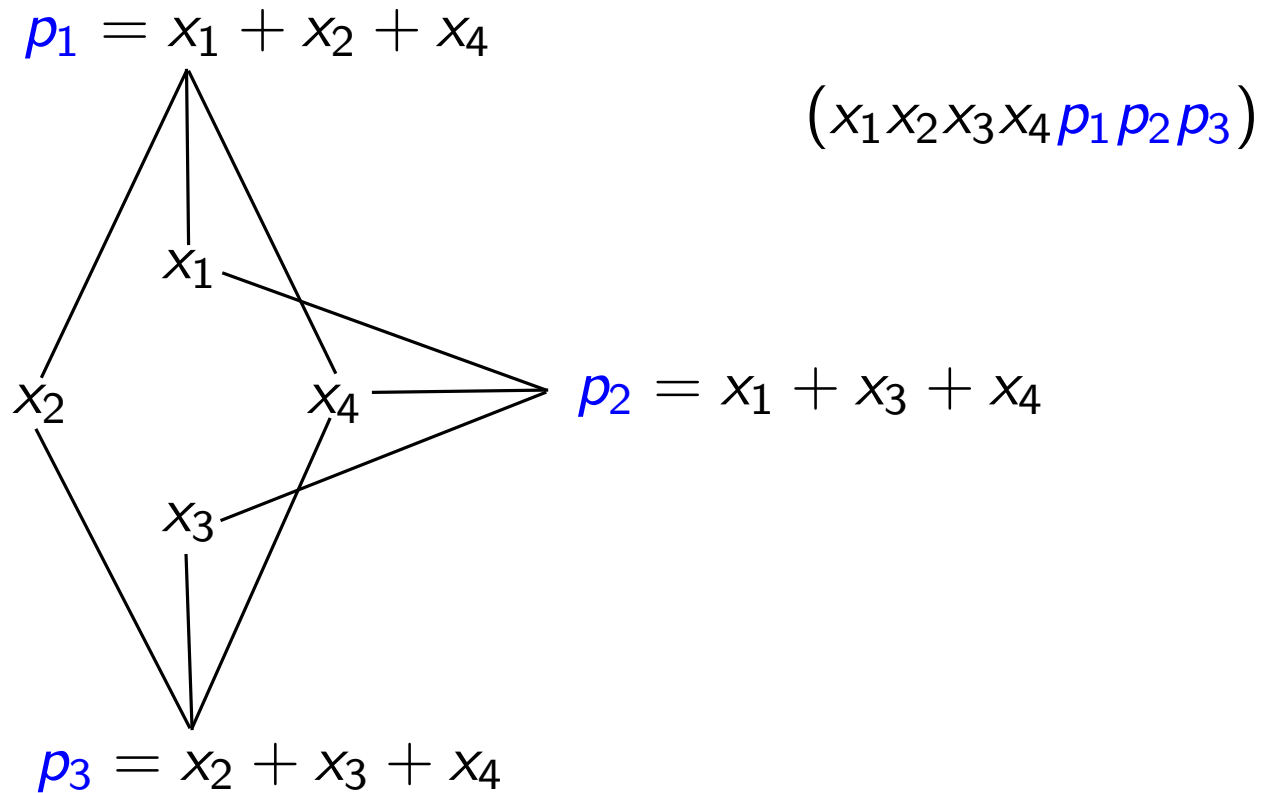
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Thus, $\text{Pr}_e \rightarrow 0$ as $m \rightarrow \infty$. However, $R \rightarrow 0$! Repetition code is **NOT** efficient! Shannon demonstrated that there exist codes which are *capacity achieving* at non-zero rates.

Hamming Code

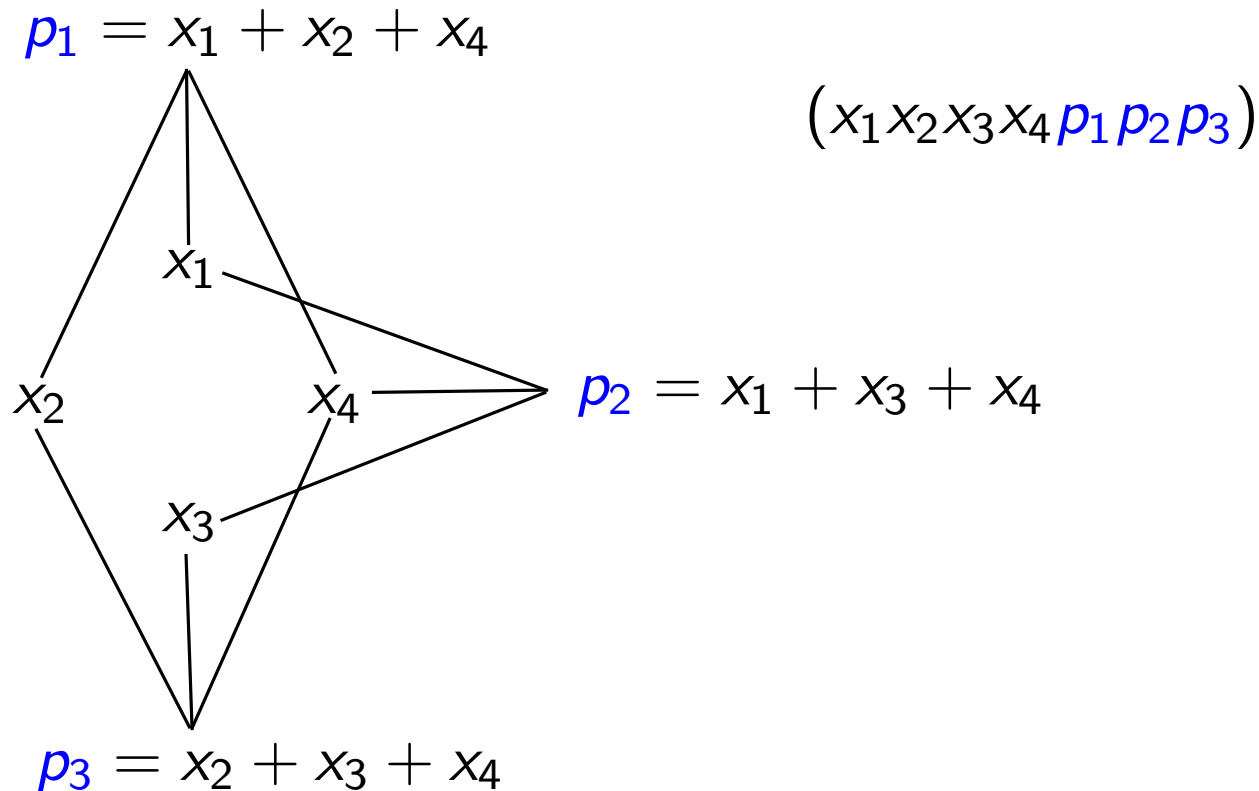


Hamming Code



The (7, 4) Hamming code can correct **1** bit error with Rate $R = 4/7$. This code is **much better** than repetition code.

Hamming Code



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Hamming codes can be computed in *linear algebra* through *matrices*. This will be explained later in this course.

Review of Probability Theory

■ Discrete Random Variables

A *discrete random variable* is used to model a “*random experiment*” with a finite or countable number of possible outcomes. For example, the toss of a coin, the roll of a die, or the count of the number of telephone calls during a given time, etc.

The *sample space* $\mathcal{S}$, of the experiment is the set of all possible outcomes and contains a finite or countable number of elements. Let $\mathcal{S} = \{\zeta_1, \zeta_2, \dots\}$.

An *event* is a subset of $\mathcal{S}$. Events consisting a *single* outcome are called *elementary events*.

Review of Probability Theory

■ Discrete Random Variables

Let X be a random variable with sample space $\mathcal{S}_X$. A *probability mass function (pmf)* for X is a mapping $p_X : \mathcal{S}_X \rightarrow [0, 1]$ from $\mathcal{S}_X$ to the closed unit interval $[0, 1]$ satisfying

$$\sum_{x \in \mathcal{S}_X} p_X(x) = 1,$$

where the number $p_X(x)$ is the *probability* that the outcome of the given random experiment is x , i.e., $p_X(x) = \Pr[X = x]$.

Review of Probability Theory

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where the number $p_X(x)$ is the *probability* that the outcome of the given random experiment is x , i.e., $p_X(x) = \Pr[X = x]$.

Every event $A \in \mathcal{S}$ has a probability $p(A) \in [0, 1]$ satisfying the following:

1. $p(A) \geq 0$
2. $p(\mathcal{S}) = 1$
3. for $A, B \in \mathcal{S}$, $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$

Review of Probability Theory

■ Discrete Random Variables

Example: A fair coin is tossed N times, and A is the event that an **even** number of heads occurs. What is $\Pr[A]$?

Review of Probability Theory

■ Discrete Random Variables

Example: A fair coin is tossed N times, and A is the event that an **even** number of heads occurs. What is $\Pr[A]$?

$$\begin{aligned}\Pr[A] &= \sum_{k=0, k \text{ even}}^N \Pr[\text{exactly } k \text{ heads occur}] \\ &= \sum_{k=0, k \text{ even}}^N \binom{N}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{N-k} \\ &= \frac{1}{2^N} \sum_{k=0, k \text{ even}}^N \binom{N}{k} \\ &= \frac{1}{2}.\end{aligned}$$

Review of Probability Theory

■ Vector Random Variables

If the elements of $\mathcal{S}_X$ are vectors of real numbers, then X is a *(real) vector random variable*.

Suppose Z is a vector random variable with a sample space in which each elements has *two* components (X, Y) , i.e.,
 $\mathcal{S}_Z = \{z_1, z_2, \dots\} = \{(x_1, y_1), (x_2, y_2), \dots\}$.

The *projection* of $\mathcal{S}_Z$ on its first coordinate is

$$\mathcal{S}_X = \{x : \text{for some } y, (x, y) \in \mathcal{S}_Z\}.$$

Review of Probability Theory

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Example: If $Z = (X, Y)$ and $\mathcal{S}_Z = \{(0, 0), (1, 0), (1, 1)\}$, then
 $\mathcal{S}_X = \mathcal{S}_Y = \{0, 1\}$.

Review of Probability Theory

■ Vector Random Variables

The *pmf* of a vector random variable $Z = (X, Y)$ is also called the *joint pmf* of X and Y , and is denoted by

$$p_Z(x, y) = p_{X,Y}(x, y) = \Pr(X = x, Y = y),$$

where the comma in the last equation denotes a logical '**AND**' operation.

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From $p_{X,Y}(x, y)$, we can find $p_X(x)$ as

$$p_X(x) \equiv p(x) = \sum_{y \in \mathcal{S}_Y} p_{X,Y}(x, y);$$

and similarly,

$$p_Y(y) \equiv p(y) = \sum_{x \in \mathcal{S}_X} p_{X,Y}(x, y).$$

Conditional Probability

- Let A and B be events, with $\Pr[A] > 0$. The *conditional probability* of B given that A occurred is

$$\Pr[B|A] = \frac{\Pr[A \cap B]}{\Pr[A]}.$$

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If $Z = (X, Y)$ and $p_X(x_k) > 0$, then

$$\begin{aligned} p_{Y|X}(y_j|x_k) &= \Pr[Y = y_j|X = x_k] \\ &= \frac{\Pr[X = x_k, Y = y_j]}{\Pr[X = x_k]} \\ &= \frac{p_{X,Y}(x_k, y_j)}{p_X(x_k)}. \end{aligned}$$

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If X and Y are *independent*, then

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x),$$

and

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x, y)}{p_X(x)} = \frac{p_X(x)p_Y(y)}{p_X(x)} = p_Y(y).$$

Expected Value

- If X is a random variable, the *expected value* (or *mean*) of X , denoted by $E[X]$, is

$$E[X] = \sum_{x \in \mathcal{S}_X} xp_X(x).$$

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In particular, $E[X^n]$ is the *n -th moment* of X . The *variance* of X is the *second* moment of $X - E[X]$, can be computed as

$$\text{VAR}[X] = E[X^2] - E[X]^2.$$

INFORMATION THEORY & CODING

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